

Morning Paper Report - Friday, September 04, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Modeling visual working memory using recurrent on-center off-surround neural network with distance dependent inhibition.

Sengupta R

Scientific reports, 2025

Source: PubMed; matched terms: visual working memory, working memory, modeling, neural network

Link: <https://pubmed.ncbi.nlm.nih.gov/40594456/>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

2. Load-Dependent Parameter Scaling in Visual Working Memory Models Reveals the Role of Decoding in Retrieval

Authors unavailable

2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, precision

Link: https://osf.io/preprints/psyarxiv/e9uys_v1/

Goal: We examined how recent models of continuous visual working memory (VWM) retrieval express capacity limitations through fitted parameters, and whether these expressions follow the inverse square-root scaling predicted by the sample-size model

Method: We examined how recent models of continuous visual working memory (VWM) retrieval express capacity limitations through fitted parameters, and whether these expressions follow the inverse square-root scaling predicted by the sample-size model

Results: We examined how recent models of continuous visual working memory (VWM) retrieval express capacity limitations through fitted parameters, and whether these expressions follow the inverse square-root scaling predicted by the sample-size model

Conclusion: Set-size related parameter changes should therefore be interpreted with consideration of the assumed decoding process, rather than as decoder-independent measures of memory precision or signal strength.

3. Neural Mechanisms of Working Memory Accuracy Revealed by Recurrent Neural Networks.

Xie Y, Liu YH, Constantinidis C, Zhou X

Frontiers in systems neuroscience, 2022

Source: PubMed; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://pubmed.ncbi.nlm.nih.gov/35237134/>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

4. Target familiarity and visual working memory do not influence familiarity effect in visual search.

Guo Z, Niu M, Wang Q

Scientific reports, 2021

Source: PubMed; matched terms: visual working memory, visual search, working memory

Link: <https://pubmed.ncbi.nlm.nih.gov/33828108/>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

5. The modulatory mechanism of spatial distance on the role of internal attention in unattended working memory representations

Authors unavailable

2024

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory representation, working memory

Link: https://osf.io/preprints/psyarxiv/cp5sr_v1/

Goal: Visual working memory (VWM) is a cognitive system that temporarily stores visual information, which is essential for higher cognitive functions

Method: Two experiments were conducted to explore how spatial distance between cued and uncued items modulates this effect

Results: The results showed that the uncued item located closer to cued items experienced less memory decline compared to those farther away, supporting the existence of an attentional spatial proximity effect

Conclusion: The results showed that the uncued item located closer to cued items experienced less memory decline compared to those farther away, supporting the existence of an attentional spatial proximity effect